

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

Based on the network traffic logs and the symptoms described, the organization experienced a Denial of Service (DoS) attack, specifically a TCP SYN flood attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The handshake consists in these 3 steps:

1. **SYN:** The client sends a SYN (Synchronize) packet to the web server to request a new connection.
2. **SYN-ACK:** The server receives the SYN packet, allocates resources for the connection, and replies with a SYN-ACK (Synchronize-Acknowledge) packet.
3. **ACK:** The client receives the SYN-ACK and responds with a final ACK (Acknowledge) packet. The connection is now established, and data can be transferred.

When a malicious actor sends a massive volume of SYN packets, they intentionally fail to complete the third step of the handshake (they never send the final ACK packet, often by spoofing their IP address). The server receives the SYN requests, replies with SYN-ACKs, and waits for the final ACKs. Because the ACKs never arrive, the server is left holding open a massive number of "half-open" connections. This rapidly consumes the server's memory, connection queues, and processing resources.

The network logs indicate an abnormally high volume of incoming TCP SYN requests originating from an unfamiliar IP address, without the corresponding ACK packets to complete the connections. This affects the server by completely exhausting its available resources. As the server becomes overwhelmed managing the malicious half-open connections, it loses its ability to respond to legitimate traffic. Consequently, regular users—such as the company's employees and customers—receive connection timeout errors, resulting in a successful Denial of Service (DoS).

Recommendations for prevention: While temporarily blocking the malicious IP address mitigated the immediate threat, it is an insufficient long-term solution against IP spoofing or Distributed Denial of Service (DDoS) attacks. To secure the network in the future, the IT

team should:

1. **Implement SYN Cookies:** Configure the server OS to use SYN cookies, which allows the server to avoid allocating resources for a connection until the final ACK is received.
2. **Intrusion Prevention System (IPS):** Deploy an IPS to automatically detect and drop traffic exhibiting abnormal patterns, such as an unusually high rate of SYN packets without completing handshakes.
3. **Rate Limiting:** Configure firewalls or edge routers to limit the number of SYN requests a single IP or subnet can make within a specific timeframe.
4. **Load Balancers / CDN:** Utilize a Content Delivery Network (CDN) or load balancers to distribute incoming traffic, which helps absorb large volumes of malicious requests before they reach the origin server.